

Centralized Multiple Moving Target Detection Using Multistatic Radar System

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Abstract—This study investigates the problem of the multiple moving target detection using multistatic radar system (MSRS). A detector based on generalized likelihood ratio test (GLRT) is presented with unknown target location and velocity. A lack of information about accurate target location will cause an error in registration. Some spatial resolution cells of no targets gather observation data of targets, which leads to extra false alarm, called “ghost target”. Therefore, “ghost target” is discriminated and the interferential amplitude is estimated by maximum likelihood estimation (MLE). Subsequently, the GLRT detector is modified to reduce the false alarm and detect target. Simulation results show that the proposed algorithm can suppress the interference while accurately detecting multiple moving targets.

Keywords—moving target detection, multistatic radar system, maximum likelihood estimation, generalized likelihood ratio test

I. INTRODUCTION

Multistatic radar system (MSRS) with spatially distributed nodes has received significant interest in recent years. Compared with the monostatic radar system, MSRS observes a common surveillance region simultaneously from different angles to utilize the spatial diversity. Various new techniques are applied to this radar configuration to improve the performance of target detection [1][2].

The problem of target detection using MSRS is determining the presence of targets in a spatial resolution cell (SRC) [3]. It is highlighted that this form of detection can be implemented in two different categories: namely “centralized” or “decentralized”. The decentralized is a two-step method in which each radar station uses its own observation data to make decisions and then fuses the local decision or statistic in the signal fusion center for further decision-making [4]–[6]. These methods behave with a performance loss because the local thresholding filters the target information. The centralized detection directly builds the global statistic using the echo of different radar sites in the fusion center.

By assuming that the location of target is a priori known, some adaptive detectors are derived in different backgrounds [7]–[9]. These algorithms just focus on such a solitary probed SRC of multistatic radar and assume other SRCs contain only noise components. The unknown parameters of target

location cause an error in echo data association. The components of target can be matched with that of noise in the relative spatial resolution cell (RSRC) [9]. These RSRCs containing no targets can be detected by mistake, which leads to an extra false alarm, that is the problem of “ghost target” [10]. The peak search method is taken to find the SRC of target within the common surveillance area [9]. However, when there are multiple targets, the strongest target will cover weak ones by the peak search. Come to the moving target detection (MTD), the detection problem becomes complex because of the unknown velocity of the target.

In this study, the multiple moving target detection algorithm is derived when there is no prior information about velocity and position of targets. First, the space mapping from the local coordinate reference to common coordinate reference is given to integrate multiple radar receiving signals. Then, in order to reject the RSRC and detect the target, the generalized likelihood ratio test (GLRT) detector is modified based on maximum likelihood estimation (MLE). Finally, the numerical experiment is given to assess the detection and “ghost target” suppression performances of the proposed algorithms.

II. RADAR AND SIGNAL MODEL

We consider an MSRS with I ($I > 1$) widely separated radars that are located at $\mathbf{p}_{R,j} = [x_{R,j}, y_{R,j}]^T$ stationarily, where $i = 0, 1, \dots, I-1$. Given that all radar sites already have time synchronization with a range cell level, they work cooperatively and beam a common surveillance region denoted by Ω at the same time.

In the surveillance area, J ($J > 0$) moving targets are located at $\mathbf{p}_{T,j}$ with the velocities $\mathbf{v}_{T,j}$, where $j = 0, 1, \dots, J-1$. All targets are assumed to facilitate the acquisition of the independent observations. Every radar transmits M ($M > 0$) pulse signals with the pulse repetition interval (PRI) T_p . The Doppler vector of the j th target in the i th radar is

$$\mathbf{d}_{i,j}(f_{d,i,j}) = [1, e^{j2\pi f_{d,i,j}T_p}, \dots, e^{j2\pi f_{d,i,j}((M-1)T_p)}]^T, \quad (1)$$

where $f_{d,i,j}$ is Doppler frequency caused by the movement between the j th target and i th radar. The time delay $\tau_{i,j} = R_{i,j}(\mathbf{p}_{T,j})/c$ is a function of target position,

where c denotes the speed of light and $R_{i,j}(\mathbf{p}_{T,j})$ is the path length from the i th radar to the j th target.

Given that all transmitted signals are observed without irrespective of range migration, the echo received by the i th radar is expressed as

$$\mathbf{R}_i = \sum_{j=0}^{J-1} \alpha_{i,j} \mathbf{d}_{i,j}^H(f_{d,i,j}) \otimes \mathbf{s}_i(\mathbf{p}_{T,j}) + \mathbf{W}_i, \quad (2)$$

$$\mathbf{s}_i(\mathbf{p}_{T,j}) \triangleq \begin{bmatrix} s_i(0 - \tau_{i,j}) \\ s_i(T_s - \tau_{i,j}) \\ \vdots \\ s_i((N-1)T_s - \tau_{i,j}) \end{bmatrix}, \mathbf{R}_i \triangleq \begin{bmatrix} \mathbf{r}_{i,0} \\ \mathbf{r}_{i,1} \\ \vdots \\ \mathbf{r}_{i,M-1} \end{bmatrix}, \quad (3)$$

where $\alpha_{i,j}$ denotes the complex scattering coefficient. T_s denotes the sample interval. N is the length of the signal, where $0 < N \leq [T_p / T_s]$. $\mathbf{W}_i \triangleq [\mathbf{w}_{i,0}, \mathbf{w}_{i,1}, \dots, \mathbf{w}_{i,M-1}]$ is the additive noise. $\mathbf{w}_{i,m}$ denotes the noise during the m th pulse. The distinct elements of the vector $\mathbf{w}_i$ is modeled as independent Gaussian random variables with zero mean and variance σ_i^2 . Due to widely spread antennas, the noise signals are assumed to be independent between different radar sites

$$E\{\mathbf{w}_{i,m} \mathbf{w}_{i',m'}^H\} = \begin{cases} \sigma_i^2 \mathbf{I}_N & i = i' \text{ and } m = m' \\ \mathbf{0}_N & \text{others} \end{cases}, \quad (4)$$

where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix, and $(\cdot)^H$ denotes the conjugate transpose operation.

III. MULTIPLE MOVING TARGET DETECTION

The problem of detecting multiple moving targets is to decide whether targets exist in common surveillance region. The common surveillance region can be partitioned into many SRCs. Given that G SRCs, which is denoted as $c(\mathbf{p}_{C,g})$ with the center $\mathbf{p}_{C,g} = [x_{C,g}, y_{C,g}]^T$, can cover the area. And each of SRC exists only one potential target, the detection problem can be modeled as testing each potential target.

The local signal is gathered based on its own coordinate reference system. To integrate multiple radar receiving signals into a single air picture, each signal of the local radar must be expressed in a common coordinate system. The space mapping is defined as

$$\mathbf{z}_i(\mathbf{p}_{C,g}) = \begin{cases} \mathbf{R}_i^H \mathbf{s}_{i,g} & |\theta(\mathbf{p}_{C,g}) - \theta_0| \leq \Delta\theta / 2 \\ 0 & |\theta(\mathbf{p}_{C,g}) - \theta_0| > \Delta\theta / 2 \end{cases}, \quad (5)$$

$$\mathbf{Z}_g \triangleq [\mathbf{z}_{0,g}, \mathbf{z}_{1,g}, \dots, \mathbf{z}_{I-1,g}], \quad (6)$$

where $\mathbf{s}_{i,g} = \mathbf{s}_i(\mathbf{p}_{C,g})$, and $\theta(\mathbf{p}_{C,g})$ represents the angle of g th SRC. θ_0 denotes the beam direction. $\Delta\theta$ is the beamwidth. $\mathbf{Z}_g$ is defined as the observation matrix of the g th potential target.

For a given SRC $c(\mathbf{p}_{C,g})$, the RSRC from the m th branch are defined as [9]

$$\Psi_{m,g} = \left\{ c(\mathbf{p}) \left| \left| \tau_m(\mathbf{p}) - \tau_m(\mathbf{p}_{C,g}) \right| < \frac{1}{2B}, \mathbf{p}_{R,g} \neq \mathbf{p}_{R,g} \right. \right\} \quad (7)$$

The relative SRCs can also gather the observation data of the target. As a result, it can be detected with a certain probability if the binary hypothesis is taken to design the detector. To solve this problem, we split the multi-targets detection problem into G ternary hypotheses

$$\begin{cases} \mathcal{H}_0: \mathbf{Z}_g = \mathbf{N}_g \\ \mathcal{H}_1: \mathbf{Z}_g = \mathbf{Y}_T + \mathbf{N}_g \\ \mathcal{H}_2: \mathbf{Z}_g = \mathbf{Y}_R + \mathbf{N}_g \end{cases}, \quad (8)$$

$$\mathbf{Y} = [\mathbf{y}_{0,g}, \mathbf{y}_{1,g}, \dots, \mathbf{y}_{I-1,g}], \quad (9)$$

$$\mathbf{N}_g = [\mathbf{n}_{0,g}, \mathbf{n}_{1,g}, \dots, \mathbf{n}_{I-1,g}], \quad (10)$$

$$\mathbf{Y}_R = [0, \dots, \mathbf{y}_{l,g}, \dots, 0], \quad (11)$$

where $\mathbf{y}_{i,g} = \alpha_{i,g} \mathbf{d}_{i,g}$, $\mathbf{n}_{l,g} \sim \mathcal{CN}(0, \sigma_i^2 \mathbf{s}_{i,g}^H \mathbf{s}_{i,g} \mathbf{I}_N)$. $\mathcal{H}_0$ and $\mathcal{H}_1$ are absent hypothesis and present hypothesis of the g th SRC respectively. Under $\mathcal{H}_2$, the SRC belongs to the RSRC.

First, the RSRC is ignored for the derivation of detector. The GLRT detector can be written as

$$L(\mathbf{Z}_g) = \frac{\sum_{l=0}^{I-1} \max_{f_{d,l,g}} |\mathbf{d}_{l,g}^H(f_{d,l,g}) \mathbf{z}_{l,g}|^2}{M^2 \sigma_i^2 \mathbf{s}_{i,g}^H \mathbf{s}_{i,g}} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\geq}} \gamma, \quad (12)$$

where the γ is threshold of constant false alarm. The statistic of RSRC can exceed the detection threshold. Thus, we modify the GLRT detector to reject the hypothesis of RSRC

$$L_m(\mathbf{Z}_g) = L(\mathbf{Z}_g) - M_R(\mathbf{Z}_g) \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\geq}} \gamma_m, \quad (13)$$

where the $M_R(\mathbf{Z}_g)$ is designed for eliminate the interference of RSRCs. γ_m is a function of false alarm. The MLE of $\mathbf{y}_{l,g}$ can be used to build the modified term $M_R(\mathbf{Z}_g)$. However, we cannot know whether the SRC belongs to RSRC. The detection performance would decline sharply if the statistic of target minus the MLE of interferences amplitude.

Thus, the SRC under test needs to be discriminated before the amplitude of interference is estimated. We designed the discriminator based on GLRT

$$\begin{aligned} D(\mathbf{Z}_g) &= \frac{\max_{\alpha_{l,g}} p(\mathbf{Z}_g | \alpha_{l,g}, \mathcal{H}_2)}{p(\mathbf{Z}_g | \mathcal{H}_0 + \mathcal{H}_1)} \\ &\propto \exp \left\{ -\frac{1}{\tilde{\sigma}_g^2} (\mathbf{z}_{l,g} - \hat{\alpha}_{l,g} \mathbf{d}_{l,g}(f_{d,l,g}))^H \right. \\ &\quad \times (\mathbf{z}_{l,g} - \hat{\alpha}_{l,g} \mathbf{d}_{l,g}(f_{d,l,g})) \\ &\quad \left. + \frac{1}{\tilde{\sigma}_g^2} \mathbf{z}_{l,g}^H \mathbf{z}_{l,g} \right\} \underset{\mathcal{H}_0 + \mathcal{H}_1}{\overset{\mathcal{H}_2}{\geq}} \eta \end{aligned}, \quad (14)$$

where η is the threshold of discriminator. $\hat{\alpha}_{l,g}$ denotes the MLE of the amplitude of interference. $\tilde{\sigma}_g^2$ represents the estimation of the noise level under $\mathcal{H}_0$ and target power under $\mathcal{H}_1$. In practice, $\tilde{\sigma}_g^2$ need to be trained using the observation vector of the probed SRC.

$$\hat{\alpha}_{l,g} = \frac{\mathbf{d}_{l,g}^H(f_{d,l,g}) \mathbf{z}_{l,g}}{M}. \quad (15)$$

By submitting (15) into (14), the discriminator can be given as

$$D(\mathbf{Z}_g) = \frac{|\mathbf{d}_{l,g}^H(f_{d,l,g}) \mathbf{z}_{l,g}|^2}{M \tilde{\sigma}_g^2} \underset{\mathcal{H}_0 + \mathcal{H}_1}{\overset{\mathcal{H}_2}{\geq}} \eta. \quad (16)$$

Given that l is an unknown parameter. Thus, (16) can be rewritten as

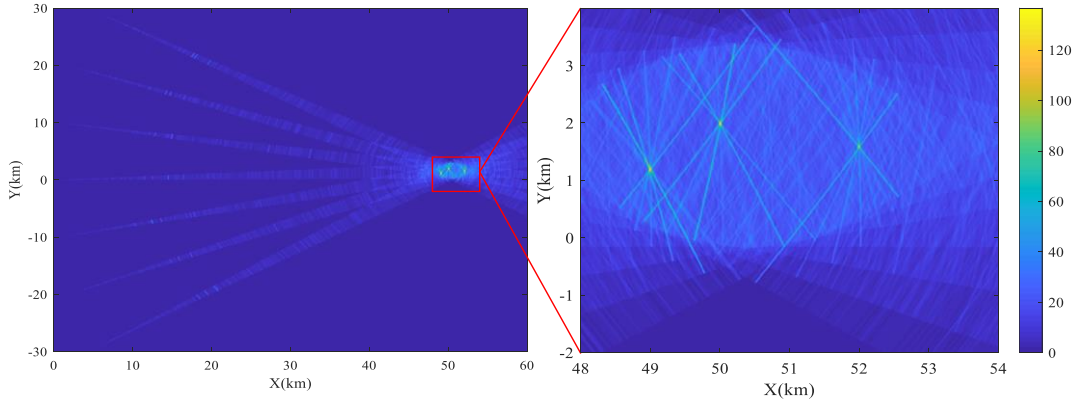


Fig. 1 Spatial power accumulation.

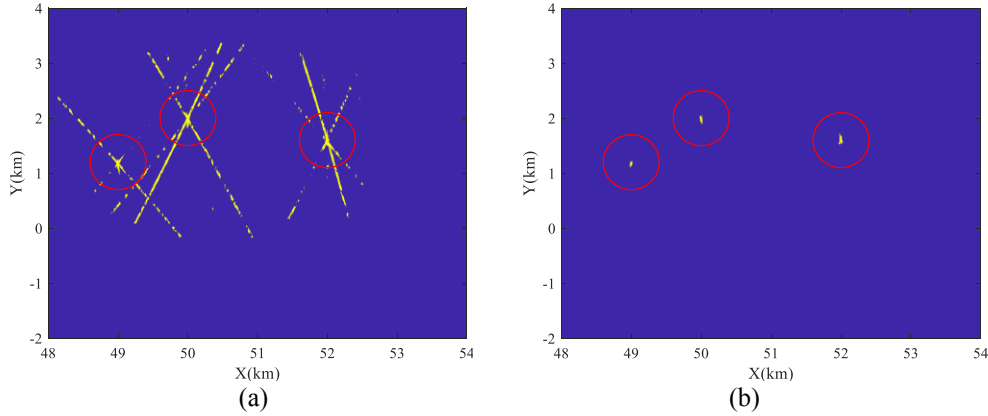


Fig. 2 Detection result.

(a) GLRT detector. (b) Modified detector.

$$D(\mathbf{Z}_g) = \max_{i,j} \frac{|\mathbf{d}_{i,g}(f_{d,i,j})\mathbf{z}_{i,g}|^2}{M\tilde{\sigma}_g^2} \underset{\mathcal{H}_0 + \mathcal{H}_1}{\overset{\mathcal{H}_2}{\geq}} \eta. \quad (17)$$

The estimated amplitude of interference is equivalent to the largest elements in the observation matrix. The threshold is η designed according to the principle of Norman-Pearson, that is

$$P_e = f_t(\eta) = P\{\mathcal{H}_2 \mid \mathcal{H}_0 + \mathcal{H}_1\}, \quad (18)$$

And then the modified term can be defined as

$$M_R(\mathbf{Z}_g) = \begin{cases} \max_{i,j} \frac{|\mathbf{d}_{i,g}(f_{d,i,j})\mathbf{z}_{i,g}|^2}{M^2\hat{\sigma}_i^2\mathbf{s}_{i,g}^H\mathbf{s}_{i,g}} & D(\mathbf{Z}_g) > \eta \\ 0 & D(\mathbf{Z}_g) < \eta \end{cases}. \quad (19)$$

IV. NUMERICAL EXPERIMENT

In this section, a scenario with three targets and a multistatic radar system is designed. The radar system involves seven radar nodes that are located at (4.4, -30), (1.8, -20), (0.8, -10), (0, 0), (0.8, 10), (1.8, 20), and (4.4, 30) km. Each radar can only receive the signal that is transmitted itself. The three targets are placed at (49, 1.2), (50, 2.0), (52, 1.6) km and move with the velocities (-100, 0), (-50, 50), (-100, 50) m/s, respectively. All radar work with the same parameters. The signal bandwidth is 1 MHz and the duration is 51 μ s. The carrier frequency is 1 GHz, the pulse number is 16, the PRI is 1ms, the beamwidth is 4°.

Fig. 1 depicts the power accumulation after spatial mapping. Power of a target echo can be integrated in its SRC.

However, the relative SRC also contains the target components. The detection results are depicted in Fig. 2, where the bright spot represents the target detected by radar system. The bright spots trapped in cycle are the targets, whereas the other spots represent the false ones. Fig. 2(a) shows that the GLRT detector generates “ghost target”, because the statistic of relative SRCs can also exceed the detection threshold with a certain probability. “ghost target” exhibits estimating numbers and position of targets after detection. In comparison, the proposed method detects all targets correctly with few “ghost targets” as shown in Fig. 2(b). And the positions of the targets can be estimated roughly, which can provide the prior information for next work. In a conclusion, the proposed method can suppress the “ghost target” while detects target correctly.

And then we compare the ROC performance of the modified detector and the GLRT detector. The proportion of the target intensity is set as 0.9:1.0:1.1 to analyze detection performance of all targets. Two measurements are taken to describe the performance of detection method, the probability of target detection and the probability of the RSRC detection. The higher probability of the RSRC detection become, the more “ghost targets” are generated.

Fig. 3 illustrates the performances of all targets at different P_e . And three targets are detected independently. Compared with the general method, the detection performance of the proposed method declines slightly, because the wrong decision in discrimination leads to an energy accumulation

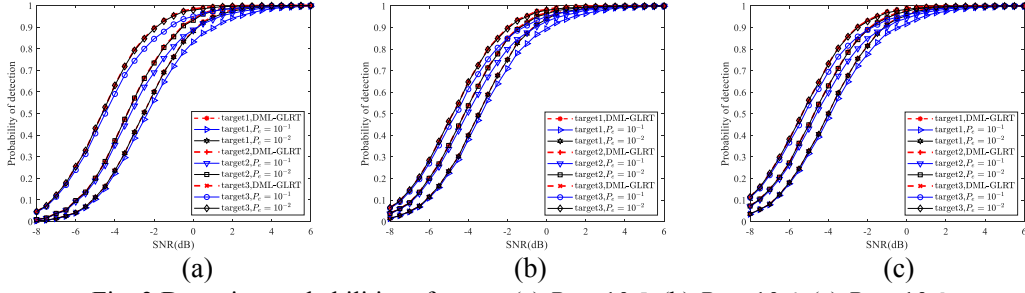


Fig. 3 Detection probabilities of target. (a) $P_{fa} = 10^{-7}$. (b) $P_{fa} = 10^{-6}$. (c) $P_{fa} = 10^{-5}$.

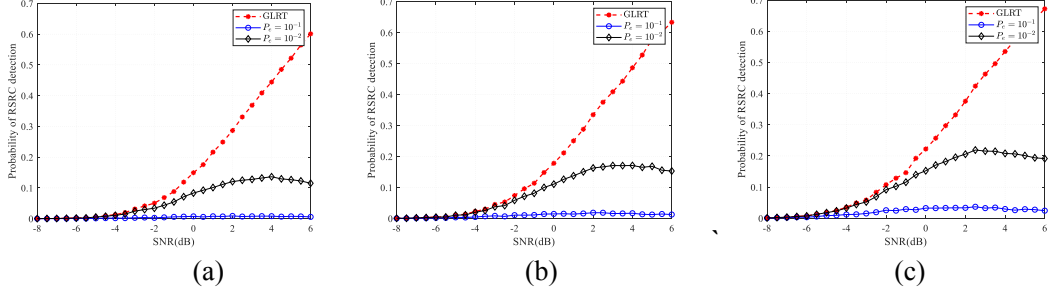


Fig. 4 Detection probabilities of the RSRC. (a) $P_{fa} = 10^{-7}$. (b) $P_{fa} = 10^{-6}$. (c) $P_{fa} = 10^{-5}$.

loss. The performance of the proposed method is the same as that of general method at $P_e = 10^{-1}$. Thus, P_e needs to be small enough to keep the good performance of target detection.

As show in Fig. 4, the detection performance of RSRC is simulated at different P_e . The higher SNR is, the more the GLRT detector generate “ghost targets”. By contrast, the proposed method maintains a low probability when the SNR is sufficiently low to be treated as noise and sufficiently high to discriminate RSRC. In Fig. 4(a), it obtains a highest probability at the SNR=4 dB where the discriminator has a bad performance. This performance degradation becomes severe in Fig. 4(b) and Fig. 4(c). The “ghost target” suppression performance is influenced by P_e and P_{fa} . As P_{fa} increases, the signal gathered in RSRC with low SNR can be detected. If the SNR is not sufficiently large to distinguish the RSRC from the target, the RSRC will be detected with a high probability. However, smaller P_e and P_f leads to a lower probability of target detection. Therefore, the P_e that matches with P_{fa} should be selected to obtain a satisfactory deception jamming suppression and target performance in practice.

V. CONCLUSION

In this study, we investigate the multiple moving target detection of MSRS when there is no prior information about target location and velocity. The space mapping which integrates the signals of different nodes into an air picture is given. The detection problem is split into test each SRC using GLRT detector in the common surveillance region. However, the RSRC can be detected by mistake, which leads to “ghost target”. In order to detect target with few false alarms, the GLRT detector is modified by eliminating the interference components in observation matrix. The simulation results show that the proposed algorithm exhibits better “ghost target” suppression performance than the

GLRT detector with a slight loss of target detection performance.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable and useful comments and suggestions which helped to improve the paper.

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